WHEN DOES A TOLERANT B-REP DEFINE A SOLID? EXISTENCE, NON-EXISTENCE AND UNIQUENESS OF REALIZATIONS

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Abstract. Boundary representations in CAD kernels carry tolerances. Edge curves, the boundary curves of the adjacent faces, and vertex points do not fit together exactly, and the solid they describe is only implied. We make this precise. A *realization* of a tolerant B-rep is a PL embedding of its cell complex whose cells stay in their tolerance zones.

Uniqueness. Suppose every shell has genus at least one, the carriers of the cells are contractible, and each shell has a product collar containing its zones and a reference surface whose inclusion is a homotopy equivalence, with the collars pairwise disjoint. Then the images of all realizations are ambient isotopic, and so are the solids they bound. The collar condition follows from a distance bound below the weak feature size of a reference surface; one step relies on Moise's theorem that homeomorphic 3-manifolds are PL homeomorphic, whose citation is still to be checked against the original. For polyhedral reference surfaces we give an explicit lower bound for the weak feature size in terms of conical radii; the key lemma is machine-checked in Lean.

Existence. Two constructions give a realization under stated hypotheses. The first uses fans in the normal disks of the edges and cones at the vertices; it is a conditional theorem whose hypotheses include two non-local ones. The second, for sharp edges with tight tolerances, re-intersects the face surfaces.

Non-existence. Under explicit conditions, if two planar patches cross transversally away from the shadows of all edge and vertex zones, there is no realization.

Experiments. OpenCascade's validity checks decide none of these questions. An ordinary sewing operation produces a wedge that passes both of OpenCascade's checks and has no realization. A thick pipe that passes both checks has two realizations whose solids are not homeomorphic.

1. Introduction

1.1. **The problem.** A solid in a CAD kernel is stored as a boundary representation: a cell complex (faces, edges, vertices) with geometry attached to each cell.

- Each face carries a trimmed patch of a surface.
- Each edge carries a 3D curve, together with one parameter-space curve (pcurve) for every adjacent face.
- Each vertex carries a point.

In floating point, and after operations such as Booleans, fillets, offsets and sewing, these data do not agree. The images of the pcurves on the surfaces miss the 3D edge curve, and the edge curves miss the vertex points. Jackson's local-tolerance approach [12], the published description of Parasolid's, attaches a tolerance to every vertex, edge and face. The data count as consistent when the deviations lie within the tolerances. OpenCascade, for instance, requires the pcurve images to lie within the edge tolerance of the 3D curve, and the edge ends within the vertex tolerance of the vertex point [15].

What solid does such a model describe? Qi and Shapiro [17] describe the conventional answer: the representation is valid if there exists an exact solid, with the same boundary combinatorial structure, in the tolerance zones. This raises two questions:

(1) *Existence.* Is there an embedded surface, with the given cell structure, that respects all the tolerances?

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(2) *Uniqueness.* If there are several, do they all bound the same solid up to ambient isotopy?

Downstream algorithms (Booleans, offsets, meshing, analysis) implicitly assume both answers are yes.

1.2. **What we prove.** We call an embedding as in (1) a *realization*: a PL embedding h of the cell complex K with $h(\operatorname{int}\sigma) \subseteq \mathcal{Z}(\sigma)$ for every cell σ, where $\mathcal{Z}(\sigma)$ is the tolerance zone of σ (Section 2).

- **Uniqueness** (Theorem 3.4).
  - *Hypotheses:* the carriers of the cells are contractible; each shell has a product collar that contains its zones and a carried reference surface whose inclusion is a homotopy equivalence; the collars are pairwise disjoint; and every shell has genus at least one.
  - *Conclusion:* the *images* of any two realizations are ambient isotopic, and so are the solids they bound.
  - *Method:* a carrier lemma (a cellular homotopy argument), combined with Waldhausen's theorem that incompressible surfaces in a product $F \times I$ can be made horizontal [23, Prop. 3.1].
- **The collar condition** (Section 4).
  - The collar condition follows from a distance bound below the weak feature size $\operatorname{wfs}(M)$ of a reference surface M (Lemma 4.1). One step relies on Moise's theorem on PL structures of 3-manifolds, whose citation is still to be checked against the original.
  - For a polyhedral M we prove $\operatorname{wfs}(M) \geq \lambda(M) > 0$, where λ is defined from the conical radii of M (Corollary 4.3). The underlying cone lemma is machine-checked in Lean 4 with Mathlib.
  - We compute lower bounds for $\operatorname{wfs}(M)$ with an octree search, up to stated floating-point qualifications.
- **Existence** (Section 5). There are two constructions:
  - fans in the normal disks of the edges, with cones at the vertices. This is a conditional theorem (Theorem 5.1), proved in Appendix A. Its hypotheses are local conditions on edges and vertices, global disjointness conditions (S1), and two non-local topological conditions (V1(b), P0).
  - re-intersection of the face surfaces, for sharp edges with tight tolerances (Propositions 5.3 and 5.4; the curved case is a sketch).
- **Non-existence** (Theorem 6.1). Under explicit conditions, if two planar patches cross transversally away from the projections ("shadows") of all edge and vertex zones, there is no realization.

1.3. **What the experiments show.** In OpenCascade 8.0.1 (Section 7):

- `BRepCheck_Analyzer` accepts models regardless of the size of the tolerances.
- `BRepAlgoAPI_Check` gives false alarms on realizable models, and accepts models whose tolerances exceed the weak feature size of the nominal surface. Among them is a thick pipe with two realizations whose solids are not homeomorphic: one ties a trefoil into the hole.
- Sewing a wedge whose faces are slightly displaced produces a model that passes both checks and, by Theorem 6.1, has no realization. The hypotheses were verified in exact rational arithmetic.

1.4. **Related work.**

Tolerant modelling.

- Segal [20] assigned a tolerance to every vertex, edge and face of polyhedral models.
- Jackson [12] gave the local-tolerance B-rep approach.
- Milenkovic [13] proved, in the plane and with a uniform ε, that normalized data have an exact realization within $\varepsilon/2$.

- Sakkalis, Shen and Patrikalakis [18] study interval solids: boxes covering the boundary of a known exact solid, up to homeomorphism. Sakkalis, Peters and Bisceglio [19, Thm. 10] prove ambient isotopy for a closed surface that lies inside a normal tube of a smooth closed surface and meets each normal fibre once. The tube radius there is bounded by curvature and by a double-normal distance.
- Qi and Shapiro [16] develop ε-solidity, which does not require an exact solid to exist [17].
- Shapiro [21] gives conditions under which the union of the tolerance zones is homotopy equivalent to the cell complex. In §5.2 (p. 178) he notes that the homotopy condition alone is not sufficient, since a valid boundary representation must also be an orientable 2-cycle separating space. It is "not immediately clear" how to extend these notions to unions of tolerance zones, though the notion of separation used in isotopy results is likely to be useful. He also treats the containment of the exact cells in their zones as a much stronger demand. Realizations satisfy exactly that containment.

Constructing an exact solid.

- Andersson, Stewart and Zidani [2] and Stewart and Zidani [22] construct a solid from inconsistent trimmed-patch data by perturbation (Whitney extension, transfinite interpolation). They *assume* that the perturbed patches embed:
  - ASZ say that adjacent patches need an a posteriori check [2, pp. 819–820];
  - SZ make a similar assumption [22, p. 188], and derive a normal-vector bound intended to exclude extraneous intersections.
- In an earlier paper [3], the same authors give checkable conditions under which a single composite parametric surface on a common parameter domain is injective, by a degree argument. They note that separately trimmed B-rep faces would need extra inside/outside information [3, §3.7].

Isotopy.

- The weak feature size and the isotopy of offsets below it are studied by Chazal and Lieutier [7] and by Chazal, Cohen-Steiner and Lieutier [5]. The latter also prove isotopy of offset level sets of general compact sets under Hausdorff perturbation [6, Thms. 4.2, 4.3].
- Chazal and Cohen-Steiner [4, Thm. 2.1] prove that a surface homeomorphic to S, lying in a thickening of S and separating its sides, is isotopic to S. This is close to the topological core of our uniqueness argument.
- Andersson, Peters and Stewart [1] treat equivalence of topological form for curvilinear objects; [we have not read its content — not yet checked against the original].

1.5. **Novelty, stated cautiously.**

- *Uniqueness.* Theorem 3.4 applies known 3-manifold topology (Waldhausen; compare [4]). What is specific is the set-up for tolerant B-reps: per-cell zones, cell-respecting realizations, the carrier lemma giving π_1-isomorphism in place of separation, and the collar obtained from wfs.
- *Weak feature size.* Positivity of wfs for compact semialgebraic sets is known: di Rocco, Eklund and Gäfvert [8, Thm. 2.5] state it and attribute it to Fu [11] [Fu's paper — not yet checked against the original]. For subanalytic sets, Chazal and Lieutier [7, Prop. 3.6] prove it for bounded open sets; the extension to compact sets is stated in their §4.3 without proof. We did not find the explicit bound $\lambda(M)$ in the literature we checked. That does not make it new.
- *Existence.* As far as we have read, the closest prior results are Milenkovic's planar existence theorem and the injectivity conditions of [3] for one composite patch. We did not find existence results for 3D B-reps with per-cell (vertex, edge, face) tolerance zones that derive the embedding of the whole cell complex from checkable hypotheses.

- *Obstruction.* We did not find a result like Theorem 6.1 in the literature we read.

2. Tolerant B-reps and realizations

2.1. **Complexes.** K is a finite CW complex whose underlying space $|K|$ is a disjoint union of closed connected surfaces $|K_1|, \dots, |K_n|$ (the *shells*). We write $\tau \le \sigma$ when τ is a face of σ (including $\tau = \sigma$). For the uniqueness theorem we assume K is *regular*: every closed cell $\bar{\sigma}$ is a ball whose boundary is a subcomplex. B-rep complexes with seam edges, closed edges or faces with holes are not regular, and must first be subdivided. A new cell inherits the zone of the open cell of K that contains it. This keeps the zones monotone, and a realization of the original model is a realization of the subdivision, since $h(\operatorname{int}\sigma') \subseteq h(\operatorname{int}\sigma) \subseteq \mathcal{Z}(\sigma)$ for a new cell $\sigma' \subseteq \sigma$. The carrier condition of Section 3 must then hold for the subdivided complex. A new edge inside a face f has carrier $\mathcal{Z}(f)$, so subdividing faces adds the hypothesis $\pi_1(\mathcal{Z}(f)) = 0$; contractible zones satisfy it. For regular K, $|K|$ carries the PL structure of the order complex of its face poset; for a B-rep complex we use the PL structure of a regular subdivision. New vertices created inside an edge or face inherit its zone, so the carrier condition requires those zones to be path-connected. The existence results say where they apply to B-rep complexes directly.

2.2. **Geometric data and zones.**

- Each vertex v has a point p_v and a tolerance τ_v.
- Each edge e has a C^2 curve $\gamma_e\colon [0,1] \to \mathbb{R}^3$ and a tolerance τ_e.
- Each face f has a C^2 surface S_f, a compact trimmed patch $P_f \subseteq S_f$, and a tolerance τ_f. The boundary of P_f consists of the images $\beta_{f,e} = S_f \circ (\text{pcurve})$ of the pcurves of the edges $e < f$.

The *deviation* of e is $\operatorname{dev}_e = \max_{f>e} \max_t |\beta_{f,e}(t) - \gamma_e(t)|$, measured at equal parameters. OpenCascade's SameParameter property asks for $\operatorname{dev}_e \le \tau_e$ in this sense; its format specification defines the edge tolerance as a one-sided distance from the pcurve images to the 3D curve [15].

Definition 2.1 (Zones)**.** We use *augmented* open zones:

- $\mathcal{Z}(v) = B(p_v, \tau_v)$;
- $\mathcal{Z}(e) = N_{\tau_e}(\gamma_e) \cup \bigcup_{v<e} \mathcal{Z}(v)$;
- $\mathcal{Z}(f) = N_{\tau_f}(P_f) \cup \bigcup_{e<f} \mathcal{Z}(e)$.

Here N_r is the open r-neighbourhood. Augmentation makes $\mathcal{Z}$ monotone: $\tau \le \sigma$ implies $\mathcal{Z}(\tau) \subseteq \mathcal{Z}(\sigma)$.

Definition 2.2 (Realization)**.** A *realization* of the tolerant B-rep $T = (K, \text{data}, \tau)$ is a PL embedding $h\colon |K| \to \mathbb{R}^3$ with $h(\operatorname{int}\sigma) \subseteq \mathcal{Z}(\sigma)$ for every cell σ.

Remark 2.3.

- *Open-cell condition.* We require only $h(\operatorname{int}\sigma) \subseteq \mathcal{Z}(\sigma)$. The closed-cell condition with unaugmented neighbourhoods, $h(\bar{\sigma}) \subseteq N_{\tau_\sigma}(\cdot)$, usually fails for every candidate, mainly because face tolerances are tiny: $h(v)$ would have to lie within τ_f of every incident patch. With augmented zones the open-cell condition implies $h(\bar{\sigma}) \subseteq \mathcal{Z}(\sigma)$, since $\bar{\sigma}$ is the union of the open cells $\tau \le \sigma$.
- *Tameness.* Requiring h to be PL rules out wild embeddings such as Alexander's horned sphere.

Definition 2.4 (Carriers)**.** The *carrier* of a cell is $\mathcal{C}(\sigma) = \bigcup_{\tau \le \sigma} \mathcal{Z}(\tau)$. For augmented zones $\mathcal{C}(\sigma) = \mathcal{Z}(\sigma)$. A continuous map $g\colon |K| \to \mathbb{R}^3$ is *carried* if $g(\bar{\sigma}) \subseteq \mathcal{C}(\sigma)$ for every σ. By Remark 2.3, every realization is carried.

Definition 2.5 (Solid)**.** Let $S_1, \dots, S_n \subset \mathbb{R}^3$ be pairwise disjoint closed surfaces. The *depth* of a point $x \notin \bigcup S_j$ is the number of j for which x lies in the bounded component of $\mathbb{R}^3 \setminus S_j$. The solid bounded by the S_j is the closure of the set of points of odd depth.

This matches the usual B-rep convention of outer shells and cavities when the shells are consistently oriented; a consistency check of the orientation is separate.

3. Uniqueness for shells of positive genus

3.1. Hypotheses.

- **Carrier condition.** $\mathcal{C}(v)$ is path-connected for every vertex v, and $\mathcal{C}(\sigma)$ is path-connected with $\pi_k(\mathcal{C}(\sigma)) = 0$ at every base point, for every k-cell σ with $k \geq 1$. Contractible carriers suffice.
- **Collar condition.** For each shell K_i there are a compact PL submanifold $U_i \subset \mathbb{R}^3$, a closed surface M_i, a PL homeomorphism $\Psi_i \colon U_i \to M_i \times [-1, 1]$, and a carried PL embedding $\varphi_i \colon |K_i| \to \mathbb{R}^3$ (a *reference*), with these properties:
  - all zones of the cells of K_i lie in $\operatorname{int} U_i$;
  - $\varphi_i(|K_i|) \subseteq \operatorname{int} U_i$, and the inclusion $\varphi_i(|K_i|) \hookrightarrow U_i$ is a homotopy equivalence;
  - the U_i are pairwise disjoint.

  (The reference need not be a level $\Psi_i^{-1}(M_i \times \{t\})$. This weaker form is what Lemma 4.1 can deliver.) Note that M_i is connected, since $M_i \simeq U_i \simeq |K_i|$; it is orientable, since $M_i \times I \cong U_i \subset \mathbb{R}^3$; and it is not a sphere when $|K_i|$ is not, since then $\pi_1(M_i) \cong \pi_1(|K_i|) \neq 1$.
- **Genus.** Each $|K_i|$ has genus at least one.

3.2. Lemmas.

Lemma 3.1 (Carrier lemma)**.** *Under the carrier condition, any two carried maps $f, g \colon |K| \to \mathbb{R}^3$ are homotopic by a homotopy H with $H(\bar{\sigma} \times I) \subseteq \mathcal{C}(\sigma)$ for every σ.*

Proof. Define H on $|K| \times \{0, 1\} \cup |K^{(k)}| \times I$ by induction on k, keeping $H(\bar{\sigma} \times I) \subseteq \mathcal{C}(\sigma)$, with $H(\cdot, 0) = f$ and $H(\cdot, 1) = g$.

- $k = 0$: for a vertex v, the points $f(v), g(v)$ lie in the path-connected set $\mathcal{C}(v)$; join them by a path in $\mathcal{C}(v)$.
- $k \geq 1$: let σ be a k-cell. $\bar{\sigma} \times I$ is a $(k+1)$-ball whose boundary $\bar{\sigma} \times \{0, 1\} \cup \partial\bar{\sigma} \times I$ is a k-sphere, and H is already defined on it:
  - on $\bar{\sigma} \times \{0, 1\}$ it is f or g, with image in $\mathcal{C}(\sigma)$;
  - on $\partial\bar{\sigma} \times I$ it is defined by induction, with image in $\bigcup_{\tau < \sigma} \mathcal{C}(\tau) \subseteq \mathcal{C}(\sigma)$.

  Since $\pi_k(\mathcal{C}(\sigma)) = 0$, H extends over $\bar{\sigma} \times I$ with image in $\mathcal{C}(\sigma)$.

The pieces agree on common boundaries, and a finite union of closed sets glued continuously gives a continuous H. □

Lemma 3.2 (Separation)**.** *Assume the carrier and collar conditions, and let h be a realization. Then $G = h(|K_i|)$ is a closed surface in $\operatorname{int} U_i$, and the inclusion $G \hookrightarrow U_i$ is a homotopy equivalence.*

Proof.

- By Lemma 3.1 on K_i, h and φ_i are homotopic through maps with image in $\bigcup \mathcal{C}(\sigma) \subseteq \operatorname{int} U_i$, so the homotopy takes place in U_i.
- $\varphi_i \colon |K_i| \to U_i$ is a homeomorphism onto $\varphi_i(|K_i|)$ followed by the inclusion, which is a homotopy equivalence by the collar condition. Hence so is $h \colon |K_i| \to U_i$.
- Since h is a homeomorphism onto G, the inclusion $G \hookrightarrow U_i$, which equals $h \circ h^{-1}$, is a homotopy equivalence.

□

Lemma 3.3 (Horizontality)**.** *Let $U \cong_\Psi M \times [-1, 1]$ be as in the collar condition, with M not a sphere, and let $G \subset \operatorname{int} U$ be a closed PL surface whose inclusion into U is a homotopy equivalence. Then there is an ambient isotopy of $\mathbb{R}^3$, supported in U and fixed on ∂U, that carries G to $\Psi^{-1}(M \times \{0\})$.*

Proof. (a) *G is incompressible.*

- By hypothesis, $\pi_1(G) \to \pi_1(U)$ is an isomorphism.
- The boundary of a compressing disk would be a curve that is essential in G and null-homotopic in U, which this rules out. That is incompressibility in the sense of Waldhausen [23, p. 58].
- $G \simeq U \simeq M$, so G is not a sphere and case (b) of that definition does not arise. We do not use the converse direction, which needs the loop theorem.
- G is a closed orientable surface in an orientable 3-manifold, hence two-sided, and it is disjoint from ∂U; so it satisfies Waldhausen's conventions [23, §0].

(b) *G can be made horizontal.*
- Apply Waldhausen's Proposition 3.1 [23, pp. 65–67] to $U \cong M \times I$ (M orientable, not a sphere) and the incompressible surface G. Since $\partial G = \emptyset$, the condition "$\partial G \subseteq F \times 1$" holds vacuously.
- It gives an isotopy, constant on ∂U, from G to a surface G' whose projection $p\colon G' \to M$ is a homeomorphism onto its image.
- $p(G')$ is open (invariance of domain) and closed in the connected surface M, so $p(G') = M$. Hence G' is the graph of a function $s\colon M \to (-1, 1)$.
- Waldhausen's proof of Proposition 3.1 is brief and uses the irreducibility of $F \times I$ [irreducibility of $F \times I$; to be checked in Hempel, ch. 10, or Moise, ch. 35 — not yet checked against the original].

(c) *Flatten.* The PL homeomorphisms of the fibres $\{x\} \times [-1, 1]$ that fix both ends and send $s(x)$ to $(1 - t)s(x)$, $t \in [0, 1]$, give an ambient isotopy of U, fixed on ∂U, from G' to $\Psi^{-1}(M \times \{0\})$. This isotopy is topological.

(d) The isotopy of (b) is an isotopy of G constant on ∂U; by isotopy extension it is induced by an ambient isotopy of U fixed on ∂U [Draft note: confirm in Waldhausen's text whether Prop. 3.1 already gives an ambient isotopy; if it is not fixed near ∂U, taper it across a collar of ∂U]. Both isotopies are fixed on ∂U and extend by the identity to $\mathbb{R}^3$.

□

3.3. **The theorem.**

Theorem 3.4 (Uniqueness)**.** *Assume the carrier condition, the collar condition, and that every shell has genus at least one. Then:*

(1) *the image of any realization is ambient isotopic (by a topological isotopy) to* $\bigcup_i \Psi_i^{-1}(M_i \times \{0\})$*;*
(2) *the images of any two realizations are ambient isotopic;*
(3) *the solids they bound (Definition 2.5) are ambient isotopic.*

Proof.
- By Lemma 3.2, each $G_i = h(|K_i|)$ satisfies the hypotheses of Lemma 3.3 in U_i. Apply that lemma shell by shell. The supports U_i are pairwise disjoint, so the isotopies Φ^i can be run simultaneously; their end map is a homeomorphism Φ of $\mathbb{R}^3$.
- $\Phi(G_j) = \Psi_j^{-1}(M_j \times \{0\})$ for every j: on U_j, every Φ^i with $i \neq j$ is the identity, because its support U_i is disjoint from U_j.
- By the Jordan–Brouwer separation theorem each closed surface separates $\mathbb{R}^3$ into a bounded and an unbounded component. A homeomorphism of $\mathbb{R}^3$ maps sets with compact closure to sets with compact closure, so Φ maps the bounded component of $\mathbb{R}^3 \setminus G_j$ onto the bounded component of $\mathbb{R}^3 \setminus \Phi(G_j)$. So depth is preserved, and Φ maps the solid bounded by the G_j onto the solid bounded by the levels.
- For two realizations with maps Φ and Φ', use $\Phi'^{-1} \circ \Phi$.

□

Remark 3.5.
- *What is not covered.* Spheres (genus 0) are excluded, and the isotopy is of images, not of embeddings.

- *Relation to Chazal–Cohen-Steiner.* Chazal and Cohen-Steiner [4, Thm. 2.1] prove that a compact orientable surface $S' \cong S$ lying in a thickening of S and separating its sides is isotopic to S in the thickening. Lemmas 3.2 and 3.3 play the same role here. What is specific to tolerant B-reps is the set-up: per-cell zones, cell-respecting realizations, the carrier lemma, and π_1 in place of separation, which the carrier lemma provides.
- *The collar condition cannot be dropped.* The thick pipe of Section 7 passes both OpenCascade checks and has two realizations whose solids are not even homeomorphic.
- *Standard facts used without separate checking against a source:* continuity of cell-by-cell extension, orientability of closed surfaces in $\mathbb{R}^3$, invariance of domain, the Jordan–Brouwer separation theorem, and that the fibrewise flattening is an ambient isotopy.

4. From the weak feature size to the collar condition

4.1. **Weak feature size.** Let $M \subset \mathbb{R}^3$ be compact, $d(p) = \operatorname{dist}(p, M)$, and let $\Gamma(p)$ be the set of nearest points of p in M. A point $p \notin M$ is *critical* if $0 \in \operatorname{conv}\{p - x : x \in \Gamma(p)\}$. The *weak feature size* $\operatorname{wfs}(M)$ is the infimum of the positive critical values of d [5, §2.2]; see also [7]. Unlike the reach, it stays positive at sharp edges and small fillets. A surface with a concave edge has reach 0, and a fillet of radius r caps the reach at r.

Lemma 4.1 (Collar)**.** *Let M be a closed connected PL surface in $\mathbb{R}^3$ of genus at least one, and let $0 < d < r < \operatorname{wfs}(M)$.*

(a) *The closed neighbourhood $\bar{N}_r(M)$ is homeomorphic to $M \times [-1, 1]$, with M going to $M \times \{0\}$.*
(a′) *There is a compact PL submanifold U with $\bar{N}_d(M) \subseteq \operatorname{int} U$, $U \subseteq \operatorname{int} \bar{N}_r(M)$ and $U \cong_{PL} M \times [-1, 1]$.*
(b) *Suppose (a′) holds. Suppose further that $M = \varphi(|K_i|)$ for a carried PL embedding φ of a shell, that every zone of a cell of that shell lies in $N_d(M)$, and that the sets $\bar{N}_r(M_i)$ of different shells are pairwise disjoint. Then the collar condition of Section 3 holds, with $U_i = U$.*

Proof. Write $d = d_M$. By the Jordan–Brouwer separation theorem, $\mathbb{R}^3 \setminus M$ has two components, a bounded one Ω_- and an unbounded one Ω_+.

(a) *Step 1: small radii.*

- d is semialgebraic (Tarski–Seidenberg), proper and nonnegative, with zero set the compact set M. So it is a *rug function* in the sense of Durfee [9, Def. 3.1].
- Work in a large simplicial cube containing $\bar{N}_r(M)$, triangulated so that M is a subcomplex. By Durfee's Propositions 3.5 and 3.7 [9], for small $\delta > 0$ there is a homeomorphism, fixing M, of $\bar{N}_\delta(M)$ onto a PL regular neighbourhood R of M.
- M is a closed orientable surface in $\mathbb{R}^3$, hence two-sided, so $R \cong M \times [-1, 1]$ with M going to $M \times \{0\}$. Hence $\bar{N}_\delta(M) \cong M \times [-1, 1]$ rel M. Its frontier $S_\delta = \{d = \delta\}$ has two components $S_\delta^\pm = S_\delta \cap \Omega_\pm$, corresponding to $M \times \{\pm 1\}$.

Step 2: the two shells. Let $A_\pm = \{x \in \Omega_\pm : \delta \le d(x) \le r\}$.

- *Inner side.* Put $O = \Omega_-$. It is bounded and open, and for $x \in O$ the distance to O^c equals $d(x)$, with the same nearest points (they lie on $\partial O = M$). So the critical points of the distance to O^c in O are exactly the critical points of d in Ω_-. The criticality notions agree: [7] use the vanishing of the generalized gradient, which holds iff $x \in \operatorname{conv} \Gamma(x)$ (see [5, §2.1]).
- Since $r < \operatorname{wfs}(M)$, the shell A_- contains no critical point. The isotopy lemma [7, Prop. 3.4] gives a homeomorphism $A_- \cong S_\delta^- \times [\delta, r]$.
- *Outer side.* Put $O = \Omega_+ \cap B(0, \mathcal{R})$ with $\mathcal{R}$ large. For $x \in O$ the distance to O^c is $\min(d(x), \mathcal{R} - |x|)$. Take $\mathcal{R}$ larger than $2r$ plus the largest $|y|$, $y \in M$. Then the points of O with distance to O^c in $[\delta, r]$ form two disjoint pieces: A_+, where the distance equals d, and a shell near the sphere $|x| = \mathcal{R}$, where it equals $\mathcal{R} - |x|$.

- Neither piece contains a critical point. In A_+ this is because $r <$ wfs. Near the sphere the only nearest point is the radial projection.
- So Prop. 3.4 gives a homeomorphism of the union with a product. Its restriction to the component A_+ gives $A_+ \cong S^+_\delta \times [\delta, r]$.

Step 3: gluing.

- A homeomorphism between manifolds with boundary maps boundary components to boundary components. So the homeomorphism of Step 2 takes $S^\pm_\delta$, a boundary component of $A_\pm$, onto one end of the product. Composing with a flip of $[\delta, r]$ if necessary, and with the inverse of its restriction to $S^\pm_\delta$, we get $A_\pm \cong S^\pm_\delta \times [0, 1]$ with $S^\pm_\delta$ going to $S^\pm_\delta \times \{0\}$.
- Then $\bar{N}_r(M) = \bar{N}_\delta(M) \cup A_+ \cup A_-$ is $M \times [-1, 1]$ with a product collar attached to each boundary component along that component. That is homeomorphic to $M \times [-2, 2] \cong M \times [-1, 1]$, with M still going to $M \times \{0\}$.

(a′) The interior of $\bar{N}_r(M)$ is $N_r(M)$: a point with $d = r$ is not critical, so d takes larger values arbitrarily near it.

- By (a), $N_r(M)$ is homeomorphic to $M \times (-1, 1)$.
- Both are PL 3-manifolds: $N_r(M)$ as an open subset of $\mathbb{R}^3$, and $M \times (-1, 1)$ with the product structure. By Moise's theorem that homeomorphic 3-manifolds are PL homeomorphic [14] [Moise, ch. 35 or 36; the noncompact case — not yet checked against the original], there is a PL homeomorphism $g\colon M \times (-1, 1) \to N_r(M)$.
- $g^{-1}(\bar{N}_d(M))$ is compact, so it lies in $M \times [-a, a]$ for some $a < 1$. Take $a < b < 1$ and $U = g(M \times [-b, b])$.
- Then U is a compact PL submanifold, $U \cong_{PL} M \times [-1, 1]$, $\bar{N}_d(M) \subseteq g(M \times (-b, b)) = \operatorname{int} U$, and $U \subseteq N_r(M) = \operatorname{int} \bar{N}_r(M)$.
- The homeomorphism g need not send M to a level, which is why the collar condition of Section 3 does not ask for that.

(b) The zones lie in $N_d(M) \subseteq \operatorname{int} U$, and $M \subseteq N_d(M) \subseteq \operatorname{int} U$. The U_i are pairwise disjoint because they lie in the disjoint sets $\bar{N}_r(M_i)$. It remains to show that the inclusion $M \hookrightarrow U$ is a homotopy equivalence.

- By (a), the composite $M \hookrightarrow U \hookrightarrow \bar{N}_r(M)$ is a homotopy equivalence. So $\pi_1(U) \to \pi_1(\bar{N}_r(M))$ is onto.
- Both groups are isomorphic to $\pi_1(M)$, a surface group of genus at least one. Surface groups are Hopfian, so a surjection from one onto an isomorphic copy is an isomorphism. Hence $\pi_1(M) \to \pi_1(U)$ is an isomorphism.
- M and $U \simeq M$ are aspherical, so by Whitehead's theorem the inclusion is a homotopy equivalence.

This uses (a) as well as (a′), and it uses the genus hypothesis: for a sphere the argument fails. Standard facts used without separate checking against a source: the Jordan–Brouwer separation theorem, Tarski–Seidenberg, regular-neighbourhood theory (the regular neighbourhood of a two-sided closed surface is a product), invariance of boundary for topological manifolds, attaching a product collar along a boundary component, that surface groups are Hopfian (they are residually finite), asphericity of closed surfaces of positive genus, and Whitehead's theorem. □

4.2. **An explicit lower bound for polyhedral surfaces.** For $a \in M$, the *conical radius* $\rho(a)$ is the supremum of the $r > 0$ such that $M \cap B(a, r) = C \cap B(a, r)$ for some closed cone C with apex a.

Lemma 4.2 (Cone lemma)**.** *Let $a \in M$, $r < \rho(a)$, and $p \notin M$ with $|p - a| < r/2$. Then every nearest point $x \in \Gamma(p)$ satisfies $\langle p - x,\, p - a\rangle = |p - x|^2$. In particular p is not critical.*

Proof. • *Nearest points lie in the ball.* $d(p) \le |p - a| < r/2$, so every $x \in \Gamma(p)$ has $|x - a| < r$.

- *The ray through x stays in M.* The points $y_s = a + s(x-a)$, for s near 1, lie in the cone and in $B(a,r)$, hence in M. So $s \mapsto |p-y_s|^2$ has a local minimum at $s=1$, which gives $\langle p-x, x-a\rangle = 0$.
- *Conclusion.* $\langle p-x, p-a\rangle = |p-x|^2 + \langle p-x, x-a\rangle = d(p)^2 > 0$ for all $x \in \Gamma(p)$. So 0 is not in the convex hull of the vectors $p-x$.

□

Corollary 4.3. *Let $\lambda(M) = \inf_{x\in M} \sup_{a\in M} \big(\rho(a)/2 - |x-a|\big)$. Then d has no critical point p with $0 < d(p) < \lambda(M)$, so $\operatorname{wfs}(M) \geq \lambda(M)$. If M is a finite union of triangles, $\lambda(M) > 0$.*

Proof.
- *The bound.* Given p with $d(p) < \lambda$ and $x \in \Gamma(p)$, pick a with $\rho(a)/2 - |x-a| > d(p)$. Then $|p-a| < \rho(a)/2$, and Lemma 4.2 applies.
- *Positivity.* For a polyhedron, $\rho(a) > 0$ for every a, by the local conical structure of polyhedra (standard; not separately checked against a source). The function $g(x) = \sup_a(\rho(a)/2 - |x-a|)$ is 1-Lipschitz and finite, since $\rho \leq \operatorname{diam} M$, and $g(x) \geq \rho(x)/2 > 0$. M is compact, so $\min g > 0$.

□

Lemma 4.2 and the pointwise form of Corollary 4.3 are machine-checked in Lean 4 with Mathlib, for subsets of an arbitrary real inner product space (Appendix B).

4.3. **A certified lower bound for** wfs**.** For a triangulated reference surface M_{PL} we compute L with "no critical value in $(c \cdot \mathrm{tol}, L)$" by a best-first octree. We call L *certified* in this qualified sense only: the computation is in double precision; containment of a point in a feature is decided with a tolerance $\mathrm{tol} = 10^{-10}$ times the model size; and the search is restricted to distances below a cut-off R.

- *Boxes near M* are certified by Lemma 4.2, using a lower bound for $\rho(a)$ at a candidate point a. The bound comes from the distance from a to the nearest feature (planar region, boundary segment, vertex) that does not contain a. It is sound for planar regions whose directed boundary segments cover their boundary, after a global coplanarity re-check of each merged region.
- *Boxes farther away* are certified by a direction test: a small linear program finds a unit vector u with $\langle u, p-x\rangle > 0$ for all candidate nearest points x of all points p of the box. The candidates are the projections of the box centre onto the relevant features.
- *Stopping rule.* The search stops at the first box that is too small to split further and cannot be certified. Its distance from M is the certified bound L.

Table 1 lists the results. Lemma 4.1 needs no critical value in $(0, r]$. The computed bound covers $(c \cdot \mathrm{tol}, L)$; the interval $(0, c \cdot \mathrm{tol}]$ is covered by Corollary 4.3 whenever $\lambda(M) > c \cdot \mathrm{tol}$, which we have not checked separately.

5. Existence

We give two constructions that turn hypotheses on the data into a realization. Neither assumes that a perturbed surface embeds as a whole. Theorem 5.1 does assume (P0) that each original patch is embedded, and two of its hypotheses are non-local. Propositions 5.3 and 5.4 assume that the re-trimmed model is embedded, which is checked computationally on the model itself.

5.1. Fans and cones.

Setting. K is regular (every face a disk bounded by a single wire). For B-rep complexes:

- at a seam edge, E3′ below refers to the two sheets of the face;
- at a closed edge, V1 counts edge ends;
- for a face with inner wires, P0 becomes "Q_f is homeomorphic to the face".

Part	exact wfs (smooth part)	certified lower bound for M_{PL}
cube, side 10	5	4.530
plate $4 \times 4 \times 1$	0.5	0.436
plate $4 \times 4 \times 0.3$	0.15	0.137
L-shape (wall 2)	1	0.894
L-shape, inner fillet 0.2	1	0.894
block with slot of width 0.2	0.1	0.086
cylinder $R = 3$, $H = 10$	3	2.512

TABLE 1. Lower bounds for wfs of the triangulated parts, certified in the qualified sense above. The exact column is the wfs of the smooth part; for a triangulated cylinder $\mathrm{wfs}(M_{PL})$ is slightly below 3. The inner fillet caps the reach at 0.2 but leaves the bound unchanged.

$\mathrm{dev}_e^*(t)$ is the deviation of the boundary points $b_{f,e}(t) = \partial P_f \cap \Pi(t)$ from $\gamma_e(t)$ in the normal plane $\Pi(t)$; it is at most dev_e.

Auxiliary sets.

- *Normal disks.* For an edge e, choose parameters $0 < a_e < b_e < 1$ and a C^1 radius function $\rho_e(t)$. ρ_e is constant on the stubs $[t_v, a_e]$, $[b_e, t_w]$ and on a neighbourhood of a_e and b_e; here t_v is the last parameter with $\gamma_e(t) \in B(p_v, r_v)$. $D_e(t)$ is the closed disk of radius $\rho_e(t)$ in $\Pi(t)$ centred at $\gamma_e(t)$, and $T_e = \bigcup_{t \in [a_e, b_e]} D_e(t)$.
- *Vertex regions.* For a vertex v, choose r_v. The vertex region W_v is the union of the ball $B(p_v, r_v)$ and, for each edge e at v, the disks $D_e(t)$ for t between t_v and a_e (or b_e).

Hypotheses. E1–E3′ are required on $[t_v, t_w]$, the stubs included.

E1: $\mathrm{dev}_e^*(t) < \rho_e(t) < \min(\tau_e, \operatorname{reach} \gamma_e|_{[a_e, b_e]})$.

E2: For every face $f > e$:

(1) $\psi_f < \pi/2$, where ψ_f is the largest angle between γ_e' and TP_f inside the tube; and $\rho_e k_f < 1$, where $k_f = \kappa_f / \cos\psi_f$;

(2) $t \mapsto \pi_e(\beta_{f,e}(t))$ is strictly monotone. A sufficient condition is $L_e + \kappa_\gamma \mathrm{dev}_e^* < 1$, where L_e is the arc-length Lipschitz constant of $\beta_{f,e} - \gamma_e$;

(3) *no return:* $\mathrm{dist}(\gamma_e(t), P_f \setminus C_f) > \rho_e(t)$. Here C_f is the union of the section arcs of P_f from $b_{f,e}$ to their exit points, of length at most $\ell_f = (\mathrm{dev}^* + \sqrt{\mathrm{dev}^{*2} + \mu\rho^2})/\mu$ with $\mu = 1 - \rho k_f$.

E3′: The exit points $k_{f,e}(t)$ and $k_{g,e}(t)$ of the two face sections on $\partial D_e(t)$ are distinct. A sufficient condition (E3) is $\alpha_e(t) > \varepsilon_f + \varepsilon_g$, with $\varepsilon = \arcsin\min\{1, (\mathrm{dev}_e^* + k\ell^2/2)/\rho_e\}$. Here α_e is the angle between the inward tangents of the two sections.

V0: γ_e's end at v and the corners of the patches at v lie in $\operatorname{int} B(p_v, r_v)$.

V1: (a) W_v is strictly star-shaped with respect to p_v, and $W_v \subseteq \operatorname{int} \mathcal{Z}(v)$. (b) *(non-local)* The construction outside W_v meets ∂W_v transversally in one simple closed curve λ_v. Each incident edge contributes one point and each incident face one arc, in the cyclic order of the link of v.

S1: (a) The W_v are pairwise disjoint. (b) The T_e are pairwise disjoint. (c) γ_e and T_e miss W_w for $w \notin \partial e$. (d) P_f misses T_e for $e \not< f$, and misses W_v for $v \not< f$. (e) Outside $\bigcup T \cup \bigcup W$, the patch pieces are pairwise disjoint. (f) $T_e \cap W_v = D_e(a_e)$ for $v < e$, and each piece outside W_v meets W_v only in λ_v.

P0: *(non-local)* Each P_f is embedded, and $P_f \cap (\bigcup T \cup \bigcup W)$ is a collar of ∂P_f, so that $Q_f = P_f \setminus \operatorname{int}(\bigcup T \cup \bigcup W)$ is a closed disk.

V1(b) and P0 are not consequences of local data. In the models of Section 7 they were verified only for the gapped cube, implicitly, through an exact check that the constructed surface is embedded. The sampled checker does not test V1(b), P0, E2.3 beyond S1(d), or S1(e) for adjacent faces.

Theorem 5.1 (Existence by fans and cones; conditional)**.** *Under E1–E3′, V0, V1, S1 and P0, the tolerant B-rep has a realization.*

The proof is in Appendix A. In outline:

- h follows γ_e along the edges and puts fans of radii in the normal disks.
- The patches, with their collars removed, are kept away from the tubes.
- Cones over the curves λ_v fill the vertex regions.
- Injectivity comes from the disjoint normal disks, E3′, star-shapedness and S1/P0.
- The resulting map is bi-Lipschitz, so a fine secant approximation is a PL realization.

Example 5.2. On a cube whose faces are pulled apart by gaps, the construction was carried out exactly in rational arithmetic: 216 triangles, with zone membership and embeddedness checked exactly (`compute/lemma0_box.py`).

5.2. **Re-intersection.**

Proposition 5.3 (Planar, degree three)**.** *Suppose every face is a disk bounded by a single wire and is planar, $P_f \subseteq \Pi_f$, and every vertex lies on exactly three faces whose planes are independent. Let c_v be the common point of the three planes, $\ell_e = [c_v, c_w]$ for an edge $e = vw$, and $\tilde{P}_f$ the polygon on the c's of the vertices of f, in the order of ∂f. Assume:*

- *$c_v \in \mathcal{Z}(v)$, $\ell_e \subseteq \mathcal{Z}(e)$ and $\tilde{P}_f \subseteq \mathcal{Z}(f)$;*
- *the $\tilde{P}_f$ are simple polygons that meet exactly in their common edges and vertices.*

Then the cellular map onto $\bigcup \tilde{P}_f$ is a realization.

Proof. It is a PL homeomorphism onto $\bigcup \tilde{P}_f$, and each cell lands in its zone. □

Proposition 5.4 (Curved, transversal)**.** *Suppose every face lies on a C^2 surface S_f, and there is a* re-trimmed model*: patches $\tilde{P}_f \subseteq S_f$, curves $\tilde{\gamma}_e$ and points c_v bounding a solid, whose cell structure is isomorphic to K (as a B-rep complex, respecting seams and closed edges). Assume:*

- *the closed cells lie in their zones;*
- *the cells are C^2 up to the boundary, the curves are regular, and there are no cusp corners;*
- *the surfaces meet at a positive angle along every non-seam edge;*
- *at every vertex the tangent cone is an embedded PL disk.*

Then a realization exists.

Proof sketch.
- Take $|K|$ to be the re-trimmed surface with its cells, and h_0 the identity.
- By compactness, $h_0(\bar{\sigma})$ has a positive margin m inside the open zone $\mathcal{Z}(\sigma)$.
- h_0 is bi-Lipschitz: within a cell by regularity, across an edge by the positive crease angle, and at a vertex by the tangent-cone condition.
- A secant approximation on a fine shape-regular curved triangulation is injective once its Lipschitz distance to h_0 is below the bi-Lipschitz constant, and it stays in the zones once its distance is below m.

□

Remark 5.5 (Checking the hypotheses).
- The re-trimmed model can be computed by extending the face surfaces and rebuilding the solid (OpenCascade's `BOPAlgo_MakerVolume`). The hypotheses are then checked on samples (`compute/reintersect_curved.py`), except the vertex part of the transversality hypothesis, which is not checked. That matters for the cylinder and the pipe, whose seam vertices lie on only two surfaces. The PL step is not carried out.

- For the planar case, the hypotheses of Proposition 5.3 for the sewn box and the sewn plate were also re-checked in exact rational arithmetic, including embeddedness.
- Results are in Section 7.

6. An obstruction

6.1. **Setting.**

- $f \neq g$ are faces whose closed cells are disks bounded by single wires.
- Their data are planar: $P_f \subseteq \Pi_f$ and $P_g \subseteq \Pi_g$, with the planes not parallel.
- $\lambda = \Pi_f \cap \Pi_g$, and $\alpha \in (0, \pi/2]$ is the angle between the planes.
- n_f, n_g are unit normals, and m_f is a unit vector in Π_f orthogonal to λ (likewise m_g).
- π_f is the orthogonal projection onto Π_f, and $U_f = \bigcup_{e<f} \mathcal{Z}(e)$, which contains the zones of the vertices of f.
- ℓ_f is the boundary loop of f: the arcs $\beta_{f,e}$ in the cyclic order of ∂f, with consecutive arcs joined by the segment between their end points.

Theorem 6.1 (Obstruction)**.** *Suppose there are $X \in \lambda$ and $\rho > 0$ such that, with $J_f = X + [-\rho, \rho]m_f$ and $J_g = X + [-\rho, \rho]m_g$:*

O0: *$\beta_{f,e} \subseteq \mathcal{Z}(e)$ for every edge e of f, and at every vertex v of f the end points of the two arcs of f at v lie in $\mathcal{Z}(v)$; likewise for g.*

O1: *$J_f \cap \ell_f = \emptyset$, and the winding number of ℓ_f in Π_f about the points of J_f is odd; likewise for g.*

O2: *$J_f \cap \operatorname{cl} \pi_f(U_f) = \emptyset$ and $J_g \cap \operatorname{cl} \pi_g(U_g) = \emptyset$.*

O3: *$\mathcal{Z}(e)$ is simply connected for every edge e of f or g.*

O4: *$\rho \sin\alpha > \tau_f \cos\alpha + \tau_g$ and $\rho \sin\alpha > \tau_g \cos\alpha + \tau_f$.*

Then the tolerant B-rep has no realization.

Proof. Suppose h is a realization, and let $\varphi = \pi_f \circ h|_{\bar{f}}$.

Step 1: over J_f, $h(\bar{f})$ lies in the slab.

- Let $y \in J_f$. We have $h(\bar{f}) \subseteq \mathcal{Z}(f) = N_{\tau_f}(P_f) \cup U_f$, and by O2 no point of U_f lies over y.
- Since $P_f \subseteq \Pi_f$, $\pi_f^{-1}(y) \cap N_{\tau_f}(P_f) \subseteq \sigma_y := y + (-\tau_f, \tau_f)n_f$.
- So $h(\bar{f}) \cap \pi_f^{-1}(y) = h(\bar{f}) \cap \sigma_y$. These points are images of interior points of f, because $h(\partial f) \subseteq U_f$.

Step 2: degree one.

- Parametrize ℓ_f over ∂f: a small arc A_v around each vertex goes to the joining segment, and the rest of each edge goes to $\beta_{f,e}$. Shrink A_v until $h(A_v) \subseteq \mathcal{Z}(v)$.
- Build a homotopy in U_f from $h|_{\partial f}$ to this parametrization:
  - on A_v, the straight-line homotopy inside the convex ball $\mathcal{Z}(v)$;
  - on the rest of an edge e, the boundary of the square is mapped into $\mathcal{Z}(e)$ (by O0, monotonicity, and $h(\bar{e}) \subseteq \mathcal{Z}(e)$), so it extends over the square by O3.
- By O2 the projected homotopy avoids every $y \in J_f$. So $\varphi|_{\partial f}$ and ℓ_f have the same winding number about y, which is odd by O1. Hence the mod-2 degree of φ at y is 1.
- Fix a triangulation of $\bar{f}$ on which h is linear. For $y \notin \varphi(\text{1-skeleton})$, which excludes only finitely many points of J_f once the slice in Step 3 is generic, $\#(h(\bar{f}) \cap \sigma_y)$ is odd.

Step 3: a crossing arc.

- Let Σ be the plane through X orthogonal to λ; it contains n_f, m_f, n_g, m_g.
- Move X along λ and shrink ρ slightly to generic values; O1, O2 and O4 are open conditions. Genericity means: Σ contains no vertex of the triangulations and meets their edges transversally, and the lines $\langle x - X, m_f \rangle = \pm\rho$, $\langle x - X, m_g \rangle = \pm\rho$ contain no vertex.
- In Σ, the rectangle $R_f = J_f + [-\tau_f, \tau_f]n_f$ meets $h(\bar{f})$ in a compact 1-manifold M_f.

- M_f misses $h(\partial f)$, since $\pi_f(R_f) = J_f$ and O2 holds. It misses the long sides of R_f, by Step 1. So its boundary lies on the two short sides.
- A generic fibre σ_y meets M_f an odd number of times (Step 2). Circle components, and arcs with both ends on one short side, meet it an even number of times.
- Hence some arc $a_f \subseteq M_f$ joins the two short sides of R_f. Likewise there is an arc a_g for g.

Step 4: the arcs meet.

- In Σ, use the coordinates $u = \langle x - X, n_f \rangle$ and $w = \langle x - X, n_g \rangle$.
- On the short sides of R_f, $|w| \geq \rho \sin\alpha - \tau_f \cos\alpha > \tau_g$ by O4, with opposite signs on the two sides. So a_f contains a sub-arc in the parallelogram $Q = \{|u| \leq \tau_f, |w| \leq \tau_g\}$ joining $w = -\tau_g$ to $w = \tau_g$. Symmetrically, a_g contains a sub-arc in Q joining $u = -\tau_f$ to $u = \tau_f$.
- By O4, Q lies in both rectangles.
- Two paths in a square joining the two pairs of opposite sides intersect [Maehara 1984, cited for this square lemma — not yet checked against the original]. So $h(\bar{f}) \cap h(\bar{g}) \neq \emptyset$ at a point that is the image of interior points of both f and g (Step 1). This contradicts injectivity.

□

Remark 6.2.

- *Which hypotheses bind.* Face tolerances are tiny (OpenCascade's default is 10^{-7}), so O4 is mild. The binding conditions are O1 and O2: the patches must cross away from the shadows of the edge and vertex zones.
- *Curved patches.* The proof uses planarity in Steps 1 and 2. A version for C^2 patches with τ_f below their reach should follow with tubular coordinates; it is not written.
- *Standard facts used without separate checking against a source:* mod-2 degree of PL maps and its relation to the winding number, general position for PL maps and generic planes, parity of crossings of a segment with a PL circle, homotopy invariance of the winding number, the nerve lemma (for the O3 test in the computations), and the square lemma cited above.

7. Experiments with OpenCascade

7.1. Set-up.

- *Software.* All experiments use OpenCascade 8.0.1 through `cadquery-ocp`. Scripts and results are in the repository [24].
- *The two checks.* "BRepCheck" is `BRepCheck_Analyzer(shape).IsValid()`. "BOP check" is `BRepAlgoAPI_Check(shape, true, true)`, which also tests geometry and self-interference.
- *Sewn models* imitate imported data. Every face of an exact part is moved by a random vector of length between $g/2$ and g, and the faces are sewn back with `BRepBuilderAPI_Sewing` at tolerance $4g$. Sewing sets each edge tolerance to 1.05 times its own deviation estimate, which is 1.048–1.05 times our measured deviation.
- *Scope.* These are a few parts in one kernel version, not a survey of industrial models.

7.2. The checks do not decide existence or uniqueness.

Table 2 summarises.

Twisted prism.

- Two non-adjacent triangular faces A, B have boundary loops with linking number -1 (by crossing count; a numerical Gauss integral agrees). The loops are $0.4037 > 2\tau = 0.1615$ apart, so the unions T_A, T_B of their edge and vertex zones are disjoint.
- In a realization, each $h(e)$ lies in the convex capsule $C_e = N_\tau(e)$, which contains the end balls (all tolerances are equal here). Homotope $h(e)$ rel end points to a segment inside C_e, and the vertices to the nominal ones inside the balls. This gives a homotopy of $h(\partial A)$ to ∂A inside T_A, so in the complement of T_B; likewise for B.

Model	BRepCheck	BOP check	What is true
twisted triangular prism, two faces with linked boundaries ($\tau = 0.081$)	valid	self-intersection	no realization (linking number -1)
thick pipe, tolerance $1.3\times$ hole radius	valid	valid	two realizations with non-homeomorphic solids (knot determinant 3 vs. 1)
sewn wedges, 30° and 10°, $g = 0.01$	valid	valid	no realization (Theorem 6.1)
sewn plate, thickness 0.05, $g = 0.01$	valid	invalid	realizable (Proposition 5.3)
filleted L-shape, $\tau = 0.1$	valid	self-intersection	$\tau +$ defl $<$ certified wfs bound of M_{PL}
long cylinder, $\tau = 3.3$	valid	valid	$\tau \geq$ wfs(nominal) (critical point exhibited)

TABLE 2. OpenCascade's checks against what the theory shows.

- The linking number is invariant under such homotopies, so $\mathrm{lk}(h(\partial A), h(\partial B)) = -1$, and $h(\partial A)$ must meet the disk $h(\bar{B})$. That contradicts embeddedness.

Knotted pipe.

- One realization keeps the nominal geometry. The other replaces the inner face by a thin tube, of radius 0.005, around a centreline that follows a long trefoil.
- The script checks the zones (maximum distance $0.983 \leq 1.3$), embeddedness ($\varepsilon\kappa_{\max} = 0.31$; non-local distance $0.036 > 2\varepsilon$ outside a window of $0.04 < \pi/\kappa_{\max} = 0.0504$) and the knot (determinant 3).
- The second solid is the exterior of a knotted arc in a ball, homeomorphic to the trefoil exterior, whose fundamental group is non-abelian [standard knot theory; textbook citation to be added — not yet checked against the original]. The nominal solid is a solid torus. So the two solids are not homeomorphic.
- This shows that the collar hypothesis of Theorem 3.4 cannot be dropped. Here $\mathrm{wfs} \leq 1 < 1.3$.

Sewn wedges.

- At the knife edge of the sewn wedges, the two planar patches cross along a line that lies outside the edge tube, by 49% (30°) and about $3\times$ (10°) of τ_e.
- The hypotheses of Theorem 6.1 hold. For the 30° wedge the largest feasible ρ at the positions searched is $1.6 \cdot 10^{-3}$, while O4 needs only $3.7 \cdot 10^{-7}$; for the 10° wedge it is $1.6 \cdot 10^{-2}$.
- An independent check in exact rational arithmetic confirmed all of O0–O4 for both wedges.
- On the sewn box and plate and on the exact box and wedge, the checker finds no crossing witness, consistent with their being realizable.

7.3. **Which construction applies.**

- *Why the fans fail on sewn data.* The fan construction fails at the vertices of sewn models. Sewing sets $\tau_e = 1.05\,\mathrm{dev}_e$, and the tube radius must exceed the deviation; but at a corner of a box three tubes of radius greater than dev do not fit disjointly into a vertex ball of radius about τ_e.
- *Re-intersection.* Re-intersection avoids this, but needs transversal faces. "Pass (curved)" refers to the edge part of the transversality hypothesis only; the vertex part is not checked.

Model	fans (Thm. 5.1)	re-intersection	margins of re-intersection (vertex / edge / face)
exact box, cylinder, pipe, slot, filleted box, 10° wedge	pass	pass where applicable	—
L-shape, all edges filleted 0.5	pass with variable radius*	not applicable	—
L-shape fused, inner fillet 0.2	fails (2 cusp vertices, 9 V1(a) failures); passes after merging coplanar faces	not applicable	—
sewn box $g = 0.01$	fails at vertices	pass	45% / 13% / 4.9%
sewn plate $g = 0.01$	fails at vertices	pass	45% / 13% / 5.9%
sewn cylinder $g = 0.01$	fails at vertices	pass (curved)	55% / 35% / 5.0%
sewn cylinder $g = 0.05$	fails at vertices	pass (curved)	55% / 11% / 8.9%
sewn pipe $g = 0.01$	fails at vertices	pass (curved)	53% / 16% / 5.1%
sewn wedges	fails at vertices	fails (edge outside tube)	no realization (Thm. 6.1)
sewn filleted box $g = 0.001$	fails at vertices	not applicable (tangent faces)	open

TABLE 3. Constructions on exact and sewn models (sampled checks, double precision). *Provisional: the checker does not yet implement the constant radius near the vertices. Face margins are the smaller of the sampled checker's value and an independent dense re-check with exact projections (`results/reintersect_B3_dense.txt`); for the sewn plate only the dense re-check is available.

- *Sewn models with tangent edges remain open.* On the sewn filleted box, OpenCascade sets each vertex tolerance equal to the largest incident edge tolerance, the patch corners lie up to $0.85\,\tau_v$ from the vertex point, and none of our vertex constructions fits.

7.4. Other observations.

- *Thickness heuristics.* Half the minimum distance between non-adjacent faces is a common estimate of the safe tolerance. It overestimates the cylinder (5 against wfs = 3), and underestimates an L-shape with a 0.05 inner fillet by a factor of about 97.
- *Offsets.* In its default mode, `BRepOffsetAPI_MakeOffsetShape` returned solids with wrong volume (errors from 44% to over a thousandfold, and in one case six shells instead of one) for a filleted box at offsets within the wfs-safe range, and BRepCheck accepted them.
- *Public documentation.* We did not find a check that tolerances leave the topology unchanged in the public documentation of OpenCascade, or in third-party documentation of ACIS and Parasolid. That says only that the documentation does not mention one.

8. Limitations and open problems

(1) **Citations behind the proofs.** Lemma 4.1(a′) uses Moise's theorem on PL structures of 3-manifolds, and Lemma 3.3 uses Waldhausen's Proposition 3.1 and the irreducibility of $F \times I$. These are marked in red until checked against the originals.

(2) **Spheres.** Theorem 3.4 excludes shells of genus 0. Waldhausen's horizontality argument does not apply to them. The PL Schoenflies theorem [Moise, ch. 17 — not yet checked against the original] should give the sphere case under the same hypotheses; this is not written.

(3) **Embeddings.** We prove that the images of realizations are ambient isotopic, not that the embeddings are isotopic as maps. The two differ by a self-homeomorphism of the surface.
(4) **Tight vertices.** We do not know whether sewn models with tangent edges and OpenCascade's vertex tolerances (equal to the largest incident edge tolerance) have realizations. Two directions are open: a vertex construction that uses the edge tubes as well as the ball, or a vertex version of the obstruction.
(5) **Mixed vertices.** Vertices where tangent and sharp edges meet need a construction combining fans and re-intersection.
(6) **Curved obstruction.** Theorem 6.1 is stated for planar patches.
(7) **Rigour of the computations.**
   - The wfs bounds and the hypothesis checks use double precision and sampling. Exceptions: the gapped-cube construction, and the exact re-verification of the sewn wedges, the sewn box and the sewn plate.
   - Interval arithmetic and a compiled implementation are needed for industrial models.
(8) **Formalization.** The carrier lemma in general (contractible carriers), Waldhausen's theorem and the existence constructions are not formalized.

[Draft note: Before submission, check against the originals:

- Moise (Geometric Topology in Dimensions 2 and 3), ch. 17 and 35: PL Schoenflies, and PL structures for Lemma 4.1(a′);
- Hempel (3-Manifolds), ch. 6 and 10: irreducibility of $F \times I$;
- Fu (1985);
- Maehara (1984);
- Andersson–Peters–Stewart (2000);
- the journal numbering of Chazal–Cohen-Steiner (2005);
- the page range of Durfee (1983);
- a textbook reference for knot exteriors.

Then revise the novelty statements in Section 1.]

Appendix A. Proof of Theorem 5.1

We use the notation and hypotheses of Section 5.1. The *exit point* $k_{f,e}(t)$ is the point where the section of P_f in $\Pi(t)$, followed from $b_{f,e}(t)$, leaves $D_e(t)$; E2 makes it unique. $m > 0$ denotes the minimum zone margin: $\rho_e < \tau_e$ and $W_v \subseteq \operatorname{int} \mathcal{Z}(v)$, and by compactness all pieces built below lie at distance at least m inside the zones of their cells.

A.1. Construction of h.

(1) $h(v) = p_v$.
(2) On $[a_e, b_e]$, $h(e) = \gamma_e$. In each $D_e(t)$, the radius $[\gamma_e(t), k_{f,e}(t)]$ is assigned to f. The fan strip is $F_{f,e} = \bigcup_{t\in[a_e,b_e]}[\gamma_e(t), k_{f,e}(t)]$.
(3) Outside the vertex regions, $h(f) = Q_f \cup \bigcup_{e<f} F_{f,e}$, where $Q_f = P_f \setminus \operatorname{int}(\bigcup T \cup \bigcup W)$.
(4) Inside W_v, take the cone from p_v over λ_v: each $x \in \lambda_v$ is joined to p_v by a segment. The cone over the point of λ_v contributed by e is $h(e)$ between v and a_e; the cone over the arc contributed by f is a sector of $h(f)$.

A.2. h is a continuous injective cellular map into the zones.

Pieces and their supports. $Q_f \subseteq P_f$ lies outside $\operatorname{int}(\bigcup T \cup \bigcup W)$. $F_{f,e} \subseteq T_e$. The cone sectors and edge stubs lie in W_v.

Injectivity inside T_e.

- The disks $D_e(t)$ lie in the normal planes and have radius below the reach, so they are pairwise disjoint [10]; this does not depend on ρ_e being constant.

- Inside one disk, the construction consists of the centre and two radii to the distinct points $k_{f,e}(t)$, $k_{g,e}(t)$ (E3′). They meet only at the centre.
- No other piece enters T_e. Non-incident pieces are excluded by S1(b)–(d) and S1(f). The patches of $f > e$ enter T_e only in the collar of ∂P_f (P0 and E2.3), and that collar is removed.

Injectivity inside W_v.

- The cone map $(s, x) \mapsto p_v + s(x - p_v)$, for $s \in (0, 1]$ and $x \in \lambda_v$, is injective. Each ray from p_v meets ∂W_v once (V1(a)), and λ_v is a simple curve on ∂W_v (V1(b)).
- No other piece enters W_v: S1(a), (c), (d) for non-incident pieces, and S1(f) for incident ones.

Injectivity elsewhere. Outside $\bigcup T \cup \bigcup W$ the pieces are the Q_f. They are embedded (P0) and pairwise disjoint (S1(e)).

Continuity and cell structure.

- *The pieces fit.* Q_f and $F_{f,e}$ agree along $k_{f,e}$. The fans and Q_f agree with the cone along λ_v, which is by V1(b) exactly their trace on ∂W_v.
- *Order.*
  - By P0, ∂Q_f is one circle made of the curves $k_{f,e}$ and arcs $A_{f,v}$ on the spheres ∂W_v.
  - By V1(b) there is exactly one arc per vertex of f, joining $k_{f,e}(a_e)$ to $k_{f,e'}(a_{e'})$.
  - So these pieces form a 2-regular graph isomorphic to ∂f, whose only cycle is ∂Q_f; the cyclic order is forced.
- *Exact intersections.*
  - $Q_f \cap F_{f,e} = k_{f,e}$, by E2.3 and S1(f).
  - $(Q_f \cup \text{fans}) \cap \text{sector}_{f,v} = A_{f,v}$, by star-shapedness and S1(f).
  - Gluing two disks along a boundary arc that is their whole intersection gives a disk. By induction, $h(f)$ is a closed disk with $\partial h(f) = \bigcup_{e<f} h(e)$.
- *h as a map.* Fix a homeomorphism $\partial f \to \partial h(f)$ that respects the cells, and extend it over f through a chart of the disk $h(f)$ (Alexander's trick).
  - On the part away from the vertex regions, take the extension piecewise C^1, by straightening the C^1 boundary of Q_f in the parameter domain of P_f.
  - On each cone sector, take it radial: the sector of $\bar{f}$ at v is the cone over an arc of the link, and a point at radius s over a link point y goes to $p_v + s(\phi(y) - p_v)$. Here ϕ is a C^1 parametrization of the arc $A_{f,v}$ of λ_v.

Zones.

- $Q_f \subseteq P_f \subseteq \mathcal{Z}(f)$.
- $F_{f,e} \subseteq T_e \subseteq N_\rho(\gamma_e) \subseteq \mathcal{Z}(e) \subseteq \mathcal{Z}(f)$.
- The cone parts lie in $W_v \subseteq \mathcal{Z}(v)$, which is contained in $\mathcal{Z}(e)$ and $\mathcal{Z}(f)$ for incident e and f.
- $h(\operatorname{int} e) \subseteq \gamma_e \cup W_v \subseteq \mathcal{Z}(e)$.
- All inclusions hold with margin at least m.

A.3. **The PL step.** *h is bi-Lipschitz.* h is injective on a compact set, so it suffices that it is locally bi-Lipschitz.

- *At the creases,* the opening angle is positive.
  - Along k, between fan and patch: the fan leaves k in direction $-\hat{r}$, and the patch section leaves with $c' \cdot \hat{r} > 0$ (E2), so the angle exceeds 90°.
  - Along γ_e, between two fans: E3′ and compactness.
  - Along ∂W_v, between the cone and the outside pieces: strict star-shapedness.
- *Where three or more pieces meet* ($k_{f,e}(a_e)$, $\gamma_e(a_e)$, corners of ∂W_v), the tangent cone is an embedded PL disk.
  - At $k(a_e)$: the fan and Q_f tangent sectors lie in the closed quarter-space $\{\langle \cdot, \gamma' \rangle \leq 0, \langle \cdot, \hat{r} \rangle \leq 0\}$. This uses $\rho_e'(a_e) = 0$: ρ_e is constant near a_e. The cone sectors lie inside

it, with $-(x-p_v)$ in its interior. The only shared rays are $-\hat{r}$ and the side-wall tangent, each between adjacent sectors.
  - At $\gamma(a_e)$: the fans lie in $\{\langle\cdot,\gamma'\rangle \geq 0\}$ and the sectors in $\{\langle\cdot,\gamma'\rangle \leq 0\}$. They share only the rays $\hat{r}_f \neq \hat{r}_g$ (E3′).
- *At the apex p_v, h* is homogeneous of degree one on each sector. It is bi-Lipschitz there because λ_v is transverse to the rays from p_v and $|x-p_v|$ is bounded below on λ_v.

Secant approximation. Take a shape-regular triangulation of $|K|$ with these properties:

- it refines the decomposition into pieces;
- it has the vertices of K as vertices;
- on each cone sector it is radial: its triangles there have the apex as a vertex and a chord of the link as the opposite side;
- h is C^1 on each closed triangle away from the apexes.

Let S be the secant map, linear on each triangle and equal to h at the vertices.

- *Away from the apexes.* h is C^1 on each closed triangle. By the standard estimate, $\mathrm{Lip}(S-h) \leq C\,\omega(Dh,\mathrm{mesh})/\sin\theta_{\min}$, where ω is a modulus of continuity and $\theta_{\min}$ the smallest angle. This tends to 0 with the mesh size.
- *On a radial triangle at the apex,* with base chord $[y_0,y_1]$: write $\hat{\phi}$ for the linear interpolation of ϕ on the chord. Then

$$(S-h)(p_v+s(y-p_v)) = s\,(\hat{\phi}(y)-\phi(y)),$$

  which is homogeneous of degree one.
  - Its radial derivative is $\hat{\phi}(y)-\phi(y)$, of size at most $\omega(\phi',\mathrm{mesh})\cdot\mathrm{mesh}$.
  - Its derivative along the chord, divided by the distance $s|y-p_v|$ to the apex, is at most $|\hat{\phi}'-\phi'|/|y-p_v| \leq \omega(\phi',\mathrm{mesh})/\min|y-p_v|$.
  - So $\mathrm{Lip}(S-h)\to 0$ uniformly on the cone sectors.

  This uses only that ϕ is C^1. It closes the gap noted in earlier versions, where the estimate was stated with a curvature of λ_v, which is only C^1.
- *Conclusion.* Since $\mathrm{Lip}(S-h)\to 0$ and $|S-h|\to 0$, for a fine enough mesh:
  - $\mathrm{Lip}(S-h)$ is below the bi-Lipschitz constant of h, so S is injective;
  - $|S-h| < m$, so $S(\mathrm{int}\,\sigma) \subseteq \mathcal{Z}(\sigma)$.

  S is PL, so it is a realization. □

Standard facts used without separate checking against a source: disjointness of normal disks below the reach [10], the implicit function theorem (for the C^1 dependence of $k_{f,e}$ on t), Alexander's trick, gluing disks along boundary arcs, and the secant-approximation estimate for C^1 maps on shape-regular triangulations.

Appendix B. Formal verification in Lean

The following are proved in Lean 4 with Mathlib (repository directory `lean/`). The build has no `sorry`, and the results depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`.

Lean statement	Content
`inner_eq_zero_of_conical`	at a nearest point x, $\langle p-x, x-a\rangle = 0$ (Lemma 4.2)
`inner_sub_eq_norm_sq_of_conical`	$\langle p-x, p-a\rangle = \lvert p-x\rvert^2$
`not_isCritical_of_conical`	p is not critical (Lemma 4.2)
`not_isCritical_of_conicalCover`	pointwise form of Corollary 4.3
`carrier_mono`, `carrier_eq_zone_of_mono`	carriers are monotone, and equal the zones when the zones are augmented
`carried_of_realization`, `image_closedCell_subset_carrier`	realizations are carried
`carried_straightLine`, `continuous_straightLine`	Lemma 3.1 when the carriers are convex

The statements are for an arbitrary real inner product space. Criticality is formalized as $0 \in \operatorname{conv}\{p - x : x \text{ nearest}\}$.

How the Lean statements correspond to the paper.

- The hypothesis `LocallyConicalAt M a r` is radial closure of M inside the open ball $B(a, r)$. The paper's definition of $\rho(a)$ implies it for $r < \rho(a)$; the two are not the same statement.
- Lean's `Carried` is pointwise on open cells. By `carrier_mono` it agrees with the paper's closed-cell definition.
- Positivity of $\lambda(M)$ for polyhedra (Corollary 4.3) is proved on paper only.

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